Documentation

- Youcef Harmouni's Build Guide pdf
  - → Code
  - → Working Principles of each component

- Materials:
  - → Theremin's Speaker



- Steps { using ultrasonic instead of an antenna | • YT: great scott

- Ultrasonic → distance measurement

- Concept: Theremin: turning on ultrasonic that reads distance, getting =, & turning that = to some sound
  - is done via analogWrite in the Arduino
  - since our hand have a max / min / in between
  - eg: you can lift your hand all the way → up ↘ down
  - in between

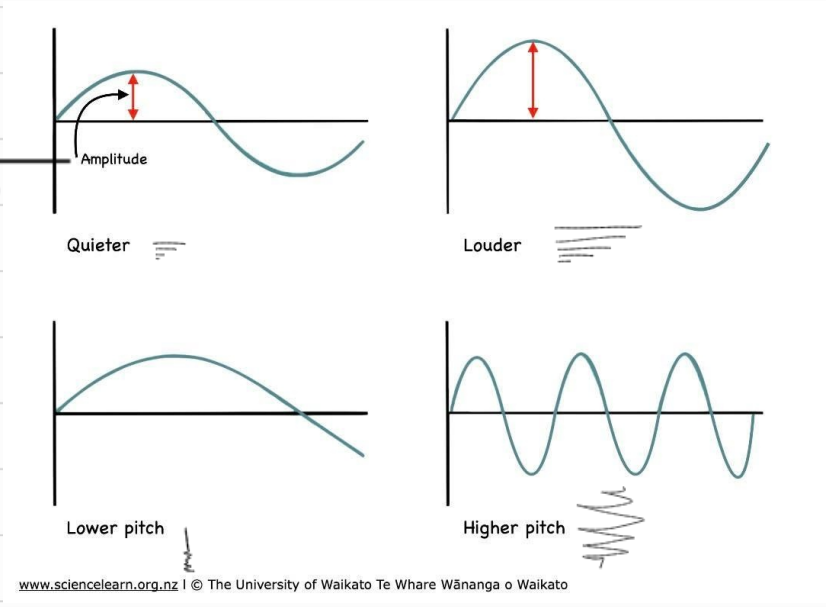
- Digital: 1/0
  Analog: every other real value
  - for the instrument sake we gonna use more values

Therefore it's Analog

- Analog values in Arduino are from 0 → 127
  Let's assume you cannot lift your hand above 50 cm
  So we need to map your 0 → 50 cm to 0 → 127 of the Arduino, and send it to the speaker (ultrasonic transducer) that emits sound waves
  And this way when you lift your hand where the ultrasonic is seeing : Distance increase ↕ / decrease ↕
  & we map it to sound, so sound get higher / lower



- Amplitude A (SI unit m):
  (distance) max displacement of a (wave) vibrating particle from its equilibrium (rest) position



- Why we need a 2nd Ultrasonic?
  The first ultrasonic can control one thing only : Volume (how loud the speaker is)
  To say we built the Theremin, we need to control another one, the Pitch

In short :

Ultrasonic x 2

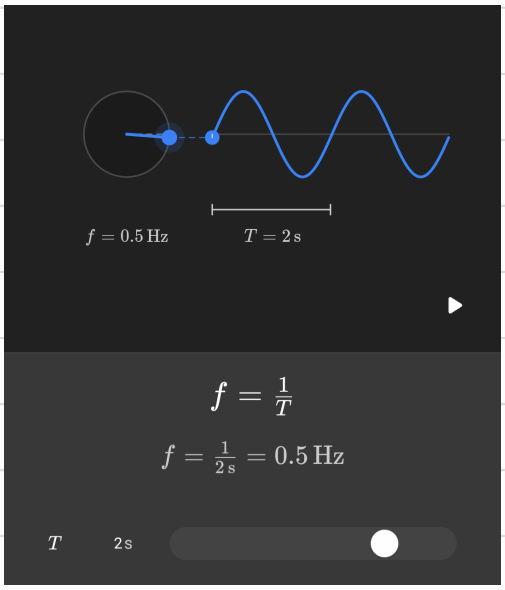
Pitch
frequency
Higher / Lower Hertz

Volume
dynamics & articulation
Amplitude

Frequency:

relates with time. We can use Arduino's delay function to simulate the pitch control

> We can make as high we want,
> But let's stick to 2K Hz



> Very first law: $T = \frac{1}{F}$
> T: period, F: frequency

T: Period: time it takes for one complete vibration

F: Frequency: how many vibrations happen in 1s, measured in Hz

—— longer delay T, fewer vibrations/s, lower frequency F
lower pitched sound

— shorter delay T, mae vibrations/s, higher frequency F
higher pitched sound

> Plan:
> Max F: delay T is 1/2000 ≈ 2 ms
> Min F: delay T is 1/200 ≈ 20 ms

} we can switch between frequencies quickly bcs the delay is short

Extra note:

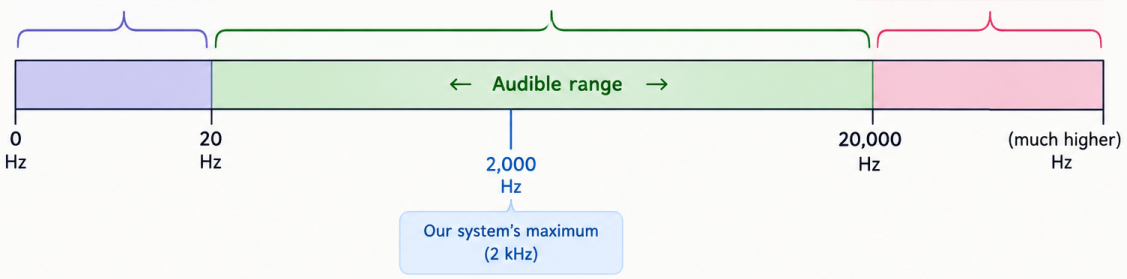
- Frequencies close in value (e.g., differing by 10 Hz) may sound very similar and be difficult to distinguish by ear (ears precieve sound proportionally)

- Frequencies below 20 Hz are inaudible, but our system's maximum is only 2000 Hz, so it won't produce inaudibly high frequencies

Infrasound (inaudible)
Frequencies below 20 Hz are generally too low for humans to hear.
(e.g. 0 – 20 Hz)

Human hearing range
Most people can hear roughly from **20 Hz** to **20,000 Hz** (20 kHz).

Ultrasound (inaudible)
Frequencies above 20,000 Hz are generally too high for humans to hear.
(e.g. > 20,000 Hz)



Why it's not a problem for higher frequencies to go inaudible?

Because our system's maximum is 2,000 Hz, which is well below 20,000 Hz (the point where sounds become inaudible).

→ So we won't accidentally generate ultrasound.

Quick numbers

- **20 Hz** → lower limit of human hearing (below this = inaudible)
- **20,000 Hz** → upper limit of human hearing (above this = inaudible)
- **2,000 Hz** → our system's maximum (still within the audible range)

Remember: Our ears are logarithmic

We don't perceive frequency changes linearly. A change from 100 Hz to 200 Hz sounds like a big difference, while 1000 Hz to 1100 Hz sounds like a much smaller change, even though both are +100 Hz.

→ We notice ratios, not fixed differences.

Summary:

- We use 2 ultrasonic
- get their distance
- Convert the ≠ to
  - volume via analogWrite
  - pitch via delay

→ We got ourselves a Theremin